A / From a code position to a goal readout

Extract selected attention-head contributions, then use the same model in a separate verbalization run.

(a) SOURCE COMPUTATION

TASK INSTRUCTIONS

Keep palindromes that start with a vowel.

CODE PREFIX

```
ys = [s for s in xs
      if s == s[::-1]]

zs =
```

t: activation at the final "=" token

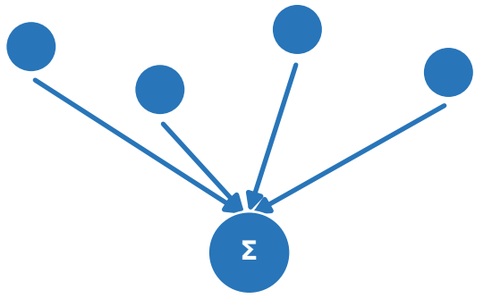
Completed: palindrome filter

Remaining: vowel-initial filter

(b) EXTRACT & AGGREGATE

Selected heads H, at token t

across source layers



$$g_t = \sum_{(\ell,h)\in H} W_O^{\ell,h}\, a_t^{\ell,h}$$

Project → sum → normalize

(c) PROBE GENERATION

FIXED TEXT PROMPT

"What are you doing at this line …?"

Source code is not in the probe prompt.

$\widehat{g}_t$

⊕ ▶ probe residual stream

⊕ ▶ probe residual stream

⊕ ▶ probe residual stream

"Keep strings starting with a vowel."

$$r_{\ell,s} \leftarrow r_{\ell,s} + \alpha \|r_{\ell,s}\| \widehat{g}_t$$

Same source vector at each selected probe layer and generation step.

DIFF readout settings: α = 0.1; layers 35–45; steering throughout decoding.

SCHEMATIC EXAMPLES · Verbalizations are illustrative, not recorded outputs. | Additive norm-relative GoalScope.